

EdgeInspect-AI

Real-time industrial defect detection and quality-control assistant for edge devices. A camera image goes through a defect detector, a severity engine and a QC advisor, and every inspection is logged, with no cloud.

Updated 09 Oct 2026, 22:41 IST Model on this page EXP-002/best.pt, 100 of 100 epochs

Stage laptop prototype

Status website <https://katyayanitiwari.github.io/edgeinspect-ai-site/>

TRAINING 100/100 epochs finished (EXP-002, Kaggle GPU)	BEST VAL MAP50 0.724 best validation epoch; test metrics are reported separately	LAST EPOCH TIME 0.1 min Kaggle Tesla T4; recorded in the run output	DATASET 1,799 images · train 1,259 / val 270 / test 270
TEST MAP50 0.770 held-out test split; EXP-002	TEST MAP50-95 0.436 held-out test split; EXP-002	LAPTOP CPU 36.6 ms median total per image; 27.3 FPS	

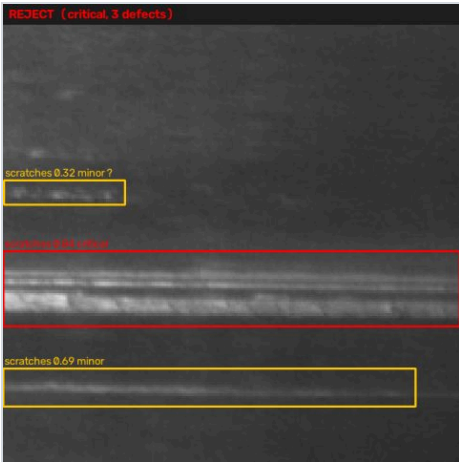
Pipeline

Six stages, in the order an image passes through them. The chip on each stage says how far it is.

Working on laptop 1 Camera / image input Image upload and webcam snapshot on the laptop. Pi Camera Module 3 later.	Training complete 2 Defect detection YOLOv8n, 6 NEU-DET classes. 100 of 100 epochs done.	Working on laptop 3 Severity engine Rules per class: box area vs train-split median and 90th percentile, plus confidence.
Placeholder 4 QC advisor Rule-based text now. Llama via ExecuTorch (required by PS5) replaces it.	Working on laptop 5 Dashboard + log Local Gradio dashboard, every inspection stored in SQLite.	Planned 6 Edge device Raspberry Pi 5 (CPU; AI HAT+ optional). Not started, hardware not yet confirmed.

Sample inspections

Real output of the current model on held-out NEU-DET test images (200×200 px steel surface). This is a trained baseline, not a production inspection system. Review false and missed detections before relying on the decisions. Uncertain detections go to manual review.

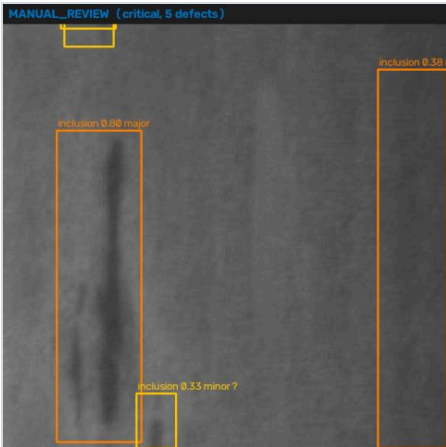


REJECT true class: scratches

DETECTED	CONF.	AREA %	SEVERITY
scratches	0.84	16.3	critical
scratches	0.69	7.38	minor
scratches	0.32	1.4	minor low conf

QC recommendation

Decision: REJECT (overall severity: critical)
Critical defect. Reject the part and quarantine it. If several rejects of the same defect type occur in a row, check the upstream process.
D01 – scratches (linear scratch marks): critical, 16.3% of image, confidence 0.84
D02 – scratches (linear scratch marks): minor, 7.38% of image, confidence 0.69
D03 – scratches (linear scratch marks): minor, 1.4% of image, confidence 0.32 - low confidence, verify



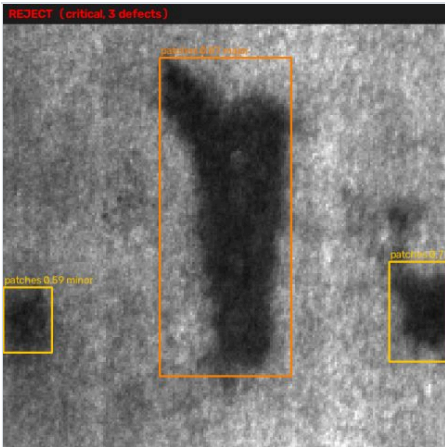
MANUAL REVIEW

true class: inclusion

DETECTED	CONF.	AREA %	SEVERITY
inclusion	0.80	13.33	major
inclusion	0.61	1.07	minor
inclusion	0.53	0.69	minor
inclusion	0.38	13.21	major low conf
inclusion	0.33	1.09	minor low conf

QC recommendation

Decision: MANUAL_REVIEW (overall severity: critical)
The model is not confident. A QC operator must inspect this part manually before a decision is made.
D01 – inclusion (foreign material embedded in the surface): major, 13.33% of image, confidence 0.80
D02 – inclusion (foreign material embedded in the surface): major, 13.21% of image, confidence 0.38 - low confidence, verify
D03 – inclusion (foreign material embedded in the surface): minor, 1.09% of image, confidence 0.33 -



REJECT true class: patches

DETECTED	CONF.	AREA %	SEVERITY
patches	0.87	21.16	major
patches	0.71	2.96	minor
patches	0.59	1.61	minor

QC recommendation

Decision: REJECT (overall severity: critical)
Critical defect. Reject the part and quarantine it. If several rejects of the same defect type occur in a row, check the upstream process.
D01 – patches (irregular surface patch): major, 21.16% of image, confidence 0.87
D02 – patches (irregular surface patch): minor, 2.96% of image, confidence 0.71
D03 – patches (irregular surface patch): minor, 1.61% of image, confidence 0.59

	low confidence, verify D04 – inclusion (foreign material embedded in the surface): minor, 1.07% of image, confidence 0.61 D05 – inclusion (foreign material embedded in the surface): minor, 0.69% of image, confidence 0.53	
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Training progress

YOLOv8n (COCO-pretrained), 320 px, SGD, up to 100 epochs with early stopping after 25 epochs without improvement. The chart shows validation metrics over 100 epochs (270 images per epoch). Final test-split metrics are shown separately above.

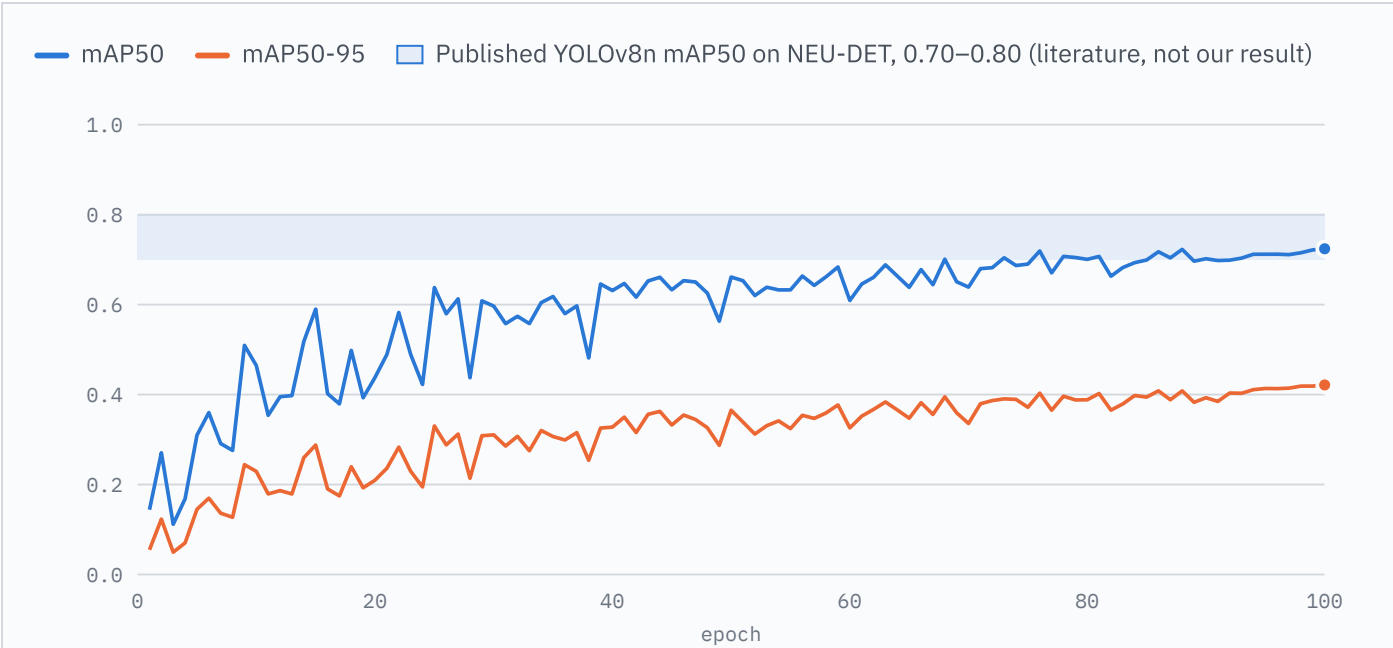


Table view

EPOCH	PRECISION	RECALL	MAP50	MAP50-95	MINUTES
1	0.006	0.561	0.144	0.055	0.1
2	0.291	0.344	0.271	0.123	0.2
3	0.221	0.203	0.112	0.049	0.3
4	0.450	0.260	0.168	0.070	0.4
5	0.437	0.420	0.310	0.145	0.5
6	0.360	0.402	0.360	0.170	0.6
7	0.334	0.337	0.291	0.136	0.7
8	0.638	0.274	0.276	0.127	0.8
9	0.482	0.538	0.509	0.244	0.9
10	0.578	0.423	0.464	0.229	1.0
11	0.431	0.329	0.354	0.179	1.1
12	0.356	0.467	0.395	0.186	1.2
13	0.468	0.417	0.398	0.179	1.3
14	0.626	0.470	0.518	0.260	1.4
15	0.526	0.631	0.590	0.288	1.5
16	0.384	0.461	0.402	0.190	1.6
17	0.433	0.450	0.379	0.175	1.7

EPOCH	PRECISION	RECALL	MAP50	MAP50-95	MINUTES
18	0.647	0.506	0.498	0.240	1.8
19	0.499	0.388	0.393	0.193	1.9
20	0.509	0.466	0.438	0.210	2.0
21	0.443	0.512	0.489	0.236	2.1
22	0.522	0.578	0.582	0.283	2.2
23	0.449	0.546	0.489	0.230	2.3
24	0.567	0.421	0.422	0.195	2.4
25	0.613	0.603	0.638	0.330	2.5
26	0.565	0.573	0.580	0.288	2.6
27	0.531	0.551	0.613	0.312	2.7
28	0.437	0.460	0.437	0.214	2.8
29	0.574	0.585	0.608	0.308	2.9
30	0.540	0.575	0.597	0.310	3.0
31	0.541	0.567	0.558	0.285	3.1
32	0.675	0.522	0.574	0.307	3.2
33	0.514	0.551	0.558	0.275	3.3
34	0.611	0.564	0.604	0.320	3.4
35	0.502	0.648	0.618	0.307	3.5
36	0.637	0.537	0.580	0.299	3.6
37	0.646	0.564	0.597	0.315	3.7
38	0.451	0.490	0.481	0.254	3.8
39	0.610	0.626	0.646	0.326	3.9
40	0.575	0.641	0.631	0.328	4.0
41	0.576	0.655	0.647	0.350	4.1
42	0.574	0.607	0.617	0.315	4.1
43	0.665	0.603	0.652	0.356	4.2
44	0.598	0.642	0.661	0.363	4.3
45	0.545	0.649	0.633	0.332	4.4
46	0.603	0.614	0.653	0.354	4.5
47	0.601	0.648	0.650	0.345	4.6
48	0.626	0.634	0.625	0.326	4.7

EPOCH	PRECISION	RECALL	MAP50	MAP50-95	MINUTES
49	0.569	0.566	0.563	0.287	4.8
50	0.596	0.645	0.661	0.365	4.9
51	0.589	0.628	0.653	0.339	5.0
52	0.635	0.557	0.620	0.312	5.1
53	0.559	0.665	0.639	0.331	5.2
54	0.581	0.620	0.633	0.341	5.3
55	0.575	0.642	0.633	0.324	5.4
56	0.546	0.671	0.664	0.354	5.5
57	0.667	0.565	0.643	0.347	5.6
58	0.628	0.652	0.662	0.359	5.7
59	0.627	0.659	0.684	0.377	5.8
60	0.581	0.619	0.609	0.326	5.9
61	0.595	0.631	0.646	0.352	6.0
62	0.595	0.668	0.661	0.367	6.1
63	0.660	0.659	0.688	0.384	6.2
64	0.557	0.678	0.663	0.366	6.3
65	0.609	0.604	0.638	0.347	6.4
66	0.658	0.620	0.678	0.382	6.5
67	0.593	0.649	0.644	0.356	6.6
68	0.592	0.688	0.701	0.395	6.7
69	0.590	0.606	0.651	0.359	6.8
70	0.628	0.585	0.639	0.336	6.9
71	0.600	0.672	0.680	0.379	7.0
72	0.673	0.621	0.682	0.387	7.0
73	0.643	0.671	0.704	0.390	7.1
74	0.608	0.692	0.687	0.389	7.2
75	0.612	0.683	0.690	0.372	7.3
76	0.709	0.669	0.719	0.403	7.4
77	0.605	0.667	0.671	0.365	7.5
78	0.634	0.686	0.707	0.396	7.6
79	0.707	0.627	0.705	0.388	7.7

EPOCH	PRECISION	RECALL	MAP50	MAP50-95	MINUTES
80	0.649	0.649	0.701	0.388	7.8
81	0.638	0.685	0.707	0.402	7.9
82	0.623	0.639	0.663	0.365	8.0
83	0.614	0.658	0.682	0.379	8.1
84	0.593	0.689	0.693	0.398	8.2
85	0.630	0.678	0.699	0.394	8.3
86	0.610	0.706	0.717	0.408	8.4
87	0.686	0.645	0.704	0.388	8.5
88	0.658	0.677	0.723	0.408	8.6
89	0.664	0.664	0.697	0.383	8.7
90	0.565	0.731	0.702	0.393	8.8
91	0.628	0.679	0.698	0.385	8.9
92	0.612	0.702	0.699	0.403	9.0
93	0.653	0.682	0.703	0.403	9.1
94	0.622	0.697	0.712	0.411	9.2
95	0.663	0.687	0.712	0.413	9.3
96	0.673	0.666	0.712	0.413	9.4
97	0.644	0.689	0.711	0.414	9.5
98	0.646	0.675	0.715	0.419	9.6
99	0.629	0.712	0.722	0.419	9.7
100	0.662	0.698	0.724	0.421	9.7

Severity engine

Each detected defect is graded by its size compared with defects of the same class in the training data. These thresholds are a data-based starting point, not an industrial standard; a real product would use its own QC specification.

CLASS	TRAIN BOXES	MAJOR ≥ AREA %	CRITICAL ≥ AREA %
crazing	483	21.1	38.0
inclusion	691	4.1	18.0
patches	613	9.4	28.2
pitted_surface	300	55.6	97.0
rolled- in_scale	433	12.4	28.1
scratches	382	7.5	16.1

- 1. Detections below 0.25 confidence are ignored.
- 2. Box area above the class median is **major**, above the class 90th percentile is **critical**, otherwise **minor**.
- 3. The part takes the worst severity. 3 or more defects raise it one level.
- 4. A confident critical defect means **REJECT**. Any defect below 0.5 confidence sends the part to **MANUAL REVIEW**. Otherwise major means **REWORK** and minor means **PASS WITH NOTE**.

Dataset

NEU-DET hot-rolled steel strip surface defects: 1,800 images found, 1 exact duplicate removed, 0 corrupted files, 0 invalid boxes. Stratified split with seed 42. Boxes per class: crazing 689, patches 881, inclusion 1,011, pitted_surface 432, rolled-in_scale 628, scratches 548.

Work completed so far

Everything below exists in the project repository and has been run at least once.

AREA	DONE	EVIDENCE
Dataset	NEU-DET chosen, inspected, cleaned (1 duplicate removed), converted to YOLO format, stratified 70/15/15 split	scripts/prepare_neu_det.py, prep_report.json
Training	Reproducible YOLOv8n training script; 1-epoch pipeline check on laptop CPU (633.6 s/epoch)	training/train.py, experiments/SANITY
Training	EXP-001 (YOLOv8n, 640 px), EXP-002 (YOLOv8n, 320 px) and EXP-003 (YOLO11n, 640 px) each completed 100 epochs	experiments/EXP-001, EXP-002 and EXP-003
Model selection	EXP-002 selected for laptop dashboard: test mAP50 0.770, mAP50-95 0.436; CPU total 36.65 ms/image	evaluation/EXP-002/metrics.json
ONNX export	EXP-002 FP32 ONNX evaluated on 270 test images; lower mAP than PyTorch, so dashboard remains on best.pt	deployment/EXP-002_ONNX_NOTES.md, evaluation/EXP-002/onnx_metrics.json
Severity	Rule-based severity engine with thresholds from train-split box sizes; 11 tests pass	inference/severity.py, tests/test_severity.py
Inference	Image / folder / webcam inference with annotated output and JSON reports	inference/detect.py
Advisor	Rule-based QC recommendation text (placeholder with the same interface Llama will use)	inference/advisor.py
Logging	Every inspection stored in a local SQLite database	inference/inspection_log.py
Dashboard	Offline laptop dashboard: upload or webcam, detections, severity, recommendation, history and stats	app/dashboard.py
Cloud training	Kaggle GPU training set up and EXP-001 result downloaded	scripts/make_kaggle_bundle.py, scripts/kaggle_run.py
Project	Git repository, 15-day plan, experiment log, official challenge facts, progress reports	PLAN.md, EXPERIMENTS.md, CHALLENGE.md, reports/

What is left to be done

The full remaining scope for Phase 1 (submission due 30 Nov 2026). Blocked items need an input from the team before work can start.

AREA	TASK	STATUS	NEEDS
Vision model	Review test errors and labels, especially lower-AP crazing and rolled-in scale	In progress	Per-class metrics and confusion matrix are generated; inspect examples
Vision model	EXP-001 / EXP-002 / EXP-003 training and held-out evaluation complete; EXP-002 selected for laptop dashboard	Done	Revisit after target-device benchmarks
Vision model	FP32 ONNX export and test evaluation complete; FP16/INT8 and target-device verification remain	In progress	Keep PyTorch best.pt until accuracy and runtime are checked
Custom data	Collect real parts, fix lighting and camera position, capture 300 to 500 images	Not started	Parts and a camera setup
Custom data	Label images, fine-tune (EXP-004), recalibrate severity thresholds	Not started	Captured images
Llama advisor	Choose a Llama-family model that fits the target device memory	Not started	Needs verification on the device
Llama advisor	Export the model with ExecuTorch and run it on-device (required by PS5)	Not started	Model access on Hugging Face (Meta licence)
Llama advisor	Design the QC prompt and replace the rule-based advisor; measure tokens/s and memory	Not started	ExecuTorch model running
Edge device	Get Raspberry Pi 5 and Camera Module 3 (AI HAT+ optional)	Blocked	Team to confirm what hardware is owned
Edge device	Pi OS setup, install script, run the full pipeline on the Pi CPU	Not started	Hardware
Edge device	Optional: Hailo NPU path (WSL2 Ubuntu, Hailo compiler, HEF model) on AI HAT+	Not started	AI HAT+ and a Hailo developer account
Application	End-to-end live pipeline: camera, detection, severity, Llama, log	Not started	Llama advisor and hardware
Application	Dashboard on the Pi with live camera view	Not started	Hardware
Benchmarks	Latency, FPS, memory and power on the Pi (CPU, and NPU if used); measured values only	Not started	Pipeline on the Pi
Submission	Team registration completed (confirmed by project owner)	Done	Keep the registration confirmation for project records

AREA	TASK	STATUS	NEEDS
Submission	Publish the source code on GitHub	Not started	A GitHub account
Submission	Technical report and demonstration video	Not started	Results from the steps above
Submission	Phase 1 submission	Not started	Due 30 Nov 2026

Challenge dates

From the official challenge page.

REGISTRATION CLOSES 19 Oct 2026	PHASE 1 SUBMISSION 30 Nov 2026	PHASE 1 RESULTS Mid-Dec 2026	PHASE 2 POC 10 Feb 2027	FINALS Early Mar 2027
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All numbers on this page are measured by the project's own scripts, except the literature range marked as such.